



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING (24CI3201)

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TITLE OF THE PROJECT	Adaptive AI-Powered Security Incident Detection and Automated	
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Project Title:

Adaptive AI-Powered Security Incident Detection and Automated

Abstract:

The **Adaptive AI-Powered Security Incident Detection and Automated Response System** is designed to address the increasing complexity and volume of cybersecurity threats faced by modern organizations. Traditional security monitoring approaches often depend on predefined rules and manual analysis, which can result in delayed detection and response when threats evolve rapidly. The proposed system provides an intelligent and scalable Security Operations Centre (SOC) architecture capable of continuously monitoring, processing, and analyzing security events in real time.

The system uses Artificial Intelligence and Machine Learning techniques to identify anomalous activities, detect potentially malicious behavior, correlate related security events, and evaluate the severity and priority of incidents. Apache Kafka is used as the real-time event streaming and communication layer, enabling efficient processing of large volumes of security events. Redis provides high-speed storage and retrieval of temporary security states, event information, and contextual data. Docker is used for containerizing individual system components, while Kubernetes provides automated deployment, orchestration, scalability, and management of distributed services.

A key feature of the proposed system is its adaptive capability. Instead of treating every security event independently, the system continuously evaluates incoming events along with their context and relationships with previously observed activities. Based on the evolving threat situation, incident priorities and recommended response actions can be dynamically adjusted. Depending on the configured response mechanisms, appropriate actions may also be initiated automatically.

The proposed architecture is expected to support faster threat detection, reduced incident response time, improved event correlation, efficient security monitoring, and better handling of dynamic and evolving cyber threats. Its distributed and containerized design also provides scalability and flexibility for deployment in environments generating large volumes of security data.

Signature of the Guide

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